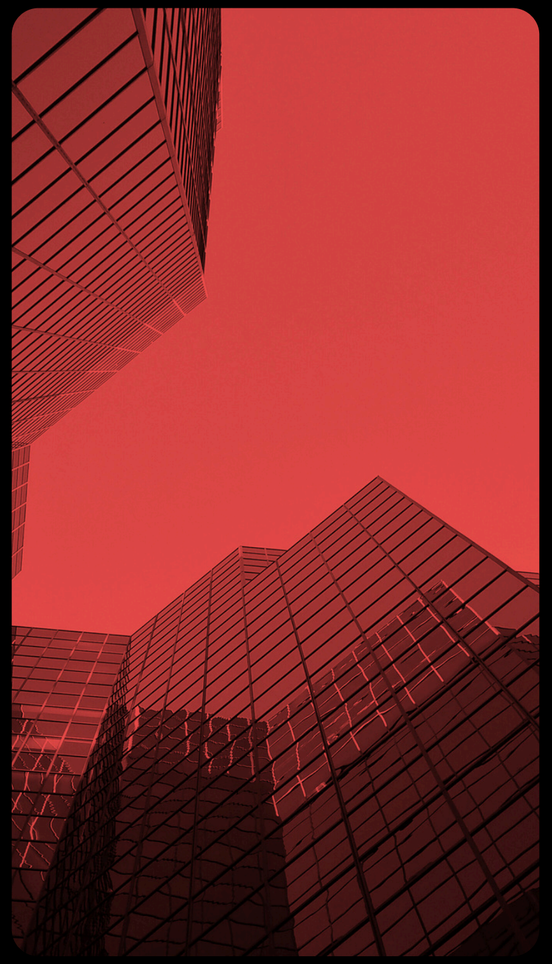


New York City Analysis

airbnb

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Our goal was to identify the economic drivers of the rental market and build a robust model to predict price categories.

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Data Cleaning

01 | **Handling Missing Data**

02 | **Outlier Removal**

03 | **Target Creation**

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Data Cleaning

HANDLING MISSING DATA

We identified that the **reviews_per_month** column had many missing values. In this context, an NA means the property received zero reviews, so we imputed these with 0 to keep the data integrity.

OUTLIER REMOVAL

We filtered the dataset to include only prices between 1 and 1000 dollars. Extreme values, like a 10,000-dollar penthouse or a 0-dollar error, would have biased our mean and ruined the model's accuracy

TARGET CREATION

Since our goal is classification, we created a binary variable: high_price. By using the Median as a threshold, I ensured that our classes are perfectly balanced (50% high, 50% low), which is essential for a fair evaluation of the model. By using the median, you force the model to truly learn the differences between the two groups to achieve a high score.

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Sampling Strategy & Logistic Regression

We used **Stratified Random Sampling** using the caret package. We split the data into:

- **Training Set** (70%) to build the model.
- **Test Set** (30%) to verify it.

Why stratified? To ensure that the proportion of 'expensive' apartments was identical in both the training and test sets, preventing biased results

We proceeded with **Logistic Regression**. We chose this specific technique because our goal had shifted to Classification, because logistic regression allows us to calculate **Odds Ratios**, giving us the ability to quantify exactly how much specific factors increase the probability of a high price.

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Stepwise Selection

```
> # Stepwise variable selection (AIC)
> logit_step <- step(logit_model, direction = "both")
Start:  AIC=29210.11
high_price ~ neighbourhood_group + room_type + number_of_reviews +
    availability_365

                     Df Deviance   AIC
<none>                     29192 29210
- number_of_reviews    1    29232 29248
- availability_365     1    29579 29595
- neighbourhood_group  4    31877 31887
- room_type            2    43354 43368
```

Once the regression framework was set, we needed to decide which variables to keep to create the most efficient model possible.

We therefore applied Stepwise Selection, guided by the AIC (Akaike Information Criterion). This process allowed us to find the statistical 'sweet spot': a model simple enough to avoid overfitting, yet powerful enough to maintain high predictive accuracy.

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```
> # Calculation of Odds Ratios (Exponential of coefficients)
> # Why: To interpret how much a variable increases the probability of a high price
> odds_ratios <- exp(coef(logit_step))
> print(odds_ratios)
                   (Intercept)     neighbourhood_groupBrooklyn     neighbourhood_groupManhattan
                    0.63869032                     3.51663846                    12.58878046
     neighbourhood_groupQueens neighbourhood_groupStaten Island             room_typePrivate room
                    1.94171799                     1.03559568                     0.04390851
          room_typeShared room                number_of_reviews                availability_365
                    0.02110169                     0.99793638                     1.00232203
```

Interpreting the Odds Ratios

Looking at the results, the "**Manhattan Factor**" is the most dominant driver: the odds ratio of 12.58 tells us that, holding all other variables constant, an apartment in Manhattan is over twelve and a half times more likely to be classified as expensive compared to the baseline location. On the other hand, the "**Room Type**" acts as a massive price reducer. The odds ratio for a Private Room is just 0.04; this indicates that listing a private room instead of an entire apartment reduces the odds of a high price by approximately 96%. These figures confirm that while other factors matter, location and privacy level are the definitive predictors of value in this dataset.

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```
> print(conf_matrix)
Confusion Matrix and Statistics

          Reference
Prediction    0    1
         0 5970 1197
         1 1368 6040

               Accuracy : 0.824
                 95% CI : (0.8177, 0.8302)
    No Information Rate : 0.5035
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.6481

 Mcnemar's Test P-Value : 0.0007889

            Sensitivity : 0.8136
            Specificity : 0.8346
         Pos Pred Value : 0.8330
         Neg Pred Value : 0.8153
             Prevalence : 0.5035
         Detection Rate : 0.4096
   Detection Prevalence : 0.4917
      Balanced Accuracy : 0.8241

       'Positive' Class : 0
```

Sensitivity and Specificity

Both values are around 81-83%. This shows that our model is equally good at identifying 'Cheap' apartments and 'Expensive' ones

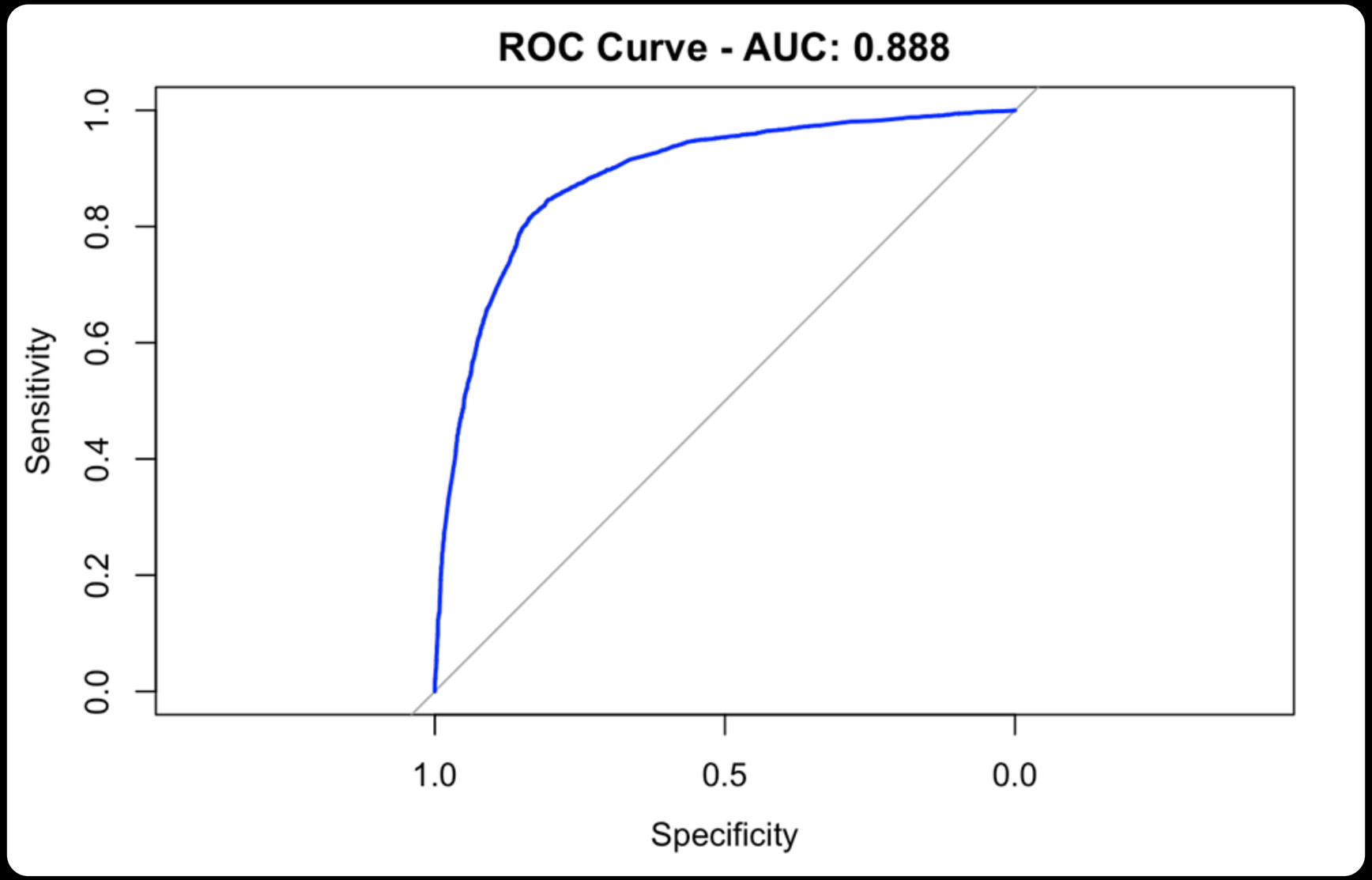
Kappa Coefficient

Our Kappa is 0.648. According to the Landis & Koch scale, any value between 0.61 and 0.80 represents a 'Substantial' agreement

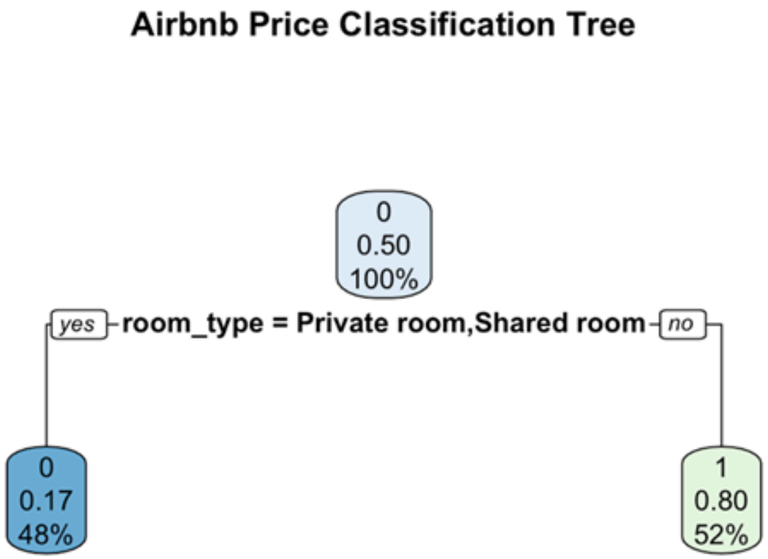
The ROC Curve

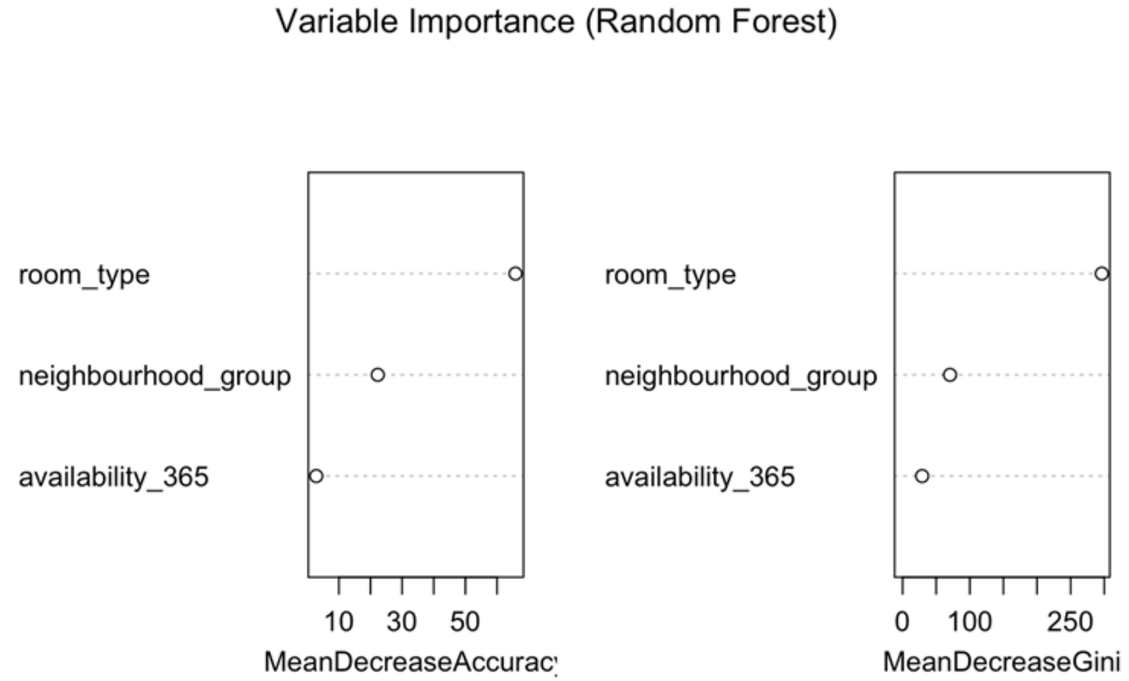
Our AUC (Area Under the Curve) is nearly 0.90. A perfect model is 1.0, and a random guess is 0.5, so our model is performing at a very high level

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The **Decision Tree** confirms that the market follows very clear rules. The first 'split', the most important one, is based on **Room Type**. If the listing is for a room, the probability of a high price is only 17%. If it is a full apartment, the probability jumps to 80%.

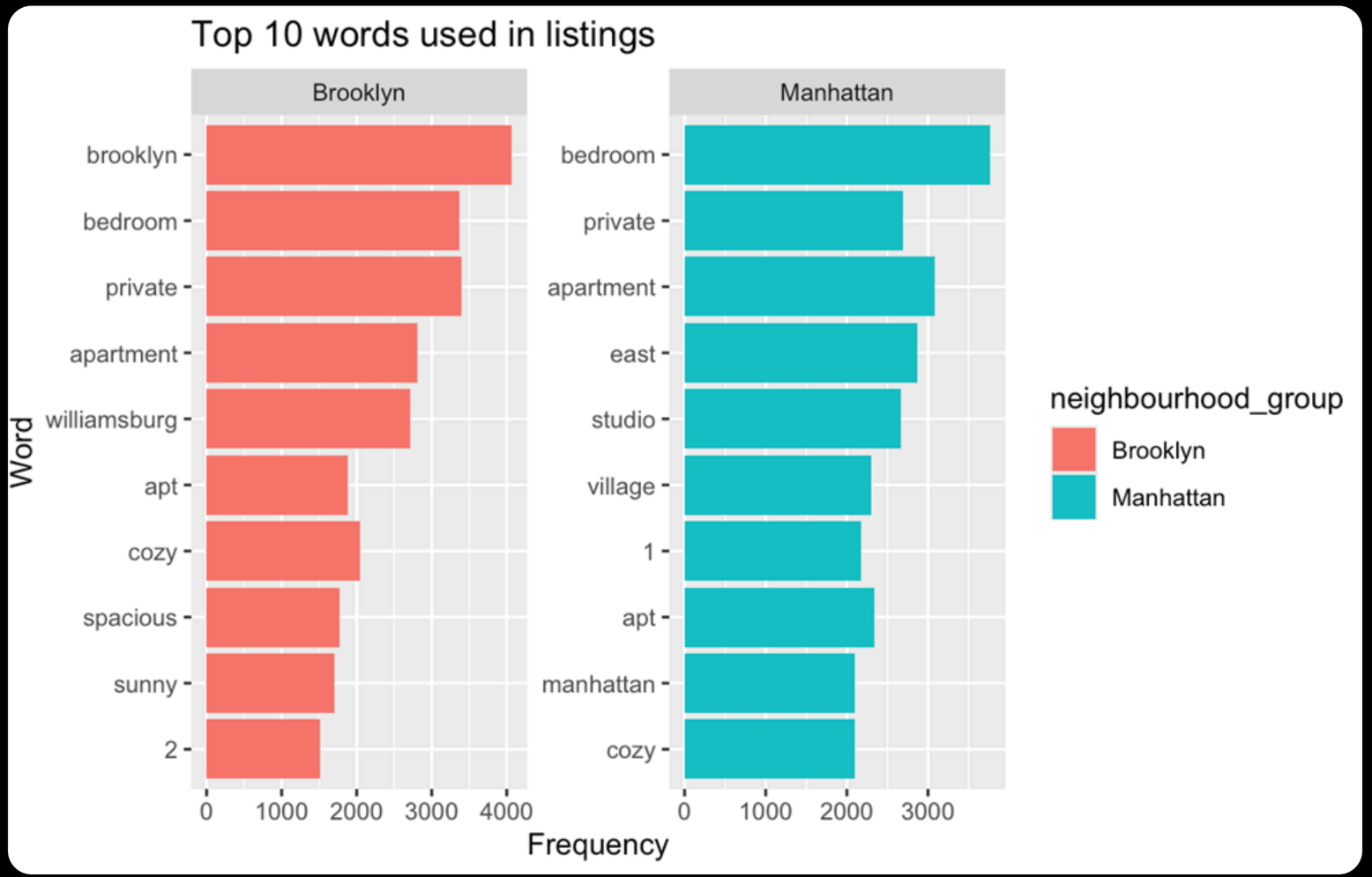
The **Random Forest** helps us rank the variables. By looking at the Mean Decrease in Accuracy, we see that **Room Type** and **Neighbourhood Group** are the dominant factors. Interestingly, the number of days an apartment is available (availability_365) is much less important for predicting the price.

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Text Mining

By comparing **Brooklyn** and **Manhattan**, we see how hosts 'sell' their property:

- Both use the word '**Cozy**' to create a feeling of comfort.
- Manhattan titles focus on efficiency and location, frequently using words like '**Studio**' and '**Village**'.
- Brooklyn hosts emphasize the physical quality of the space, using words like '**Spacious**' and '**Sunny**' to compete with the smaller apartments in Manhattan.



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Conclusion



Our analysis shows that Airbnb prices in NYC are not random. They are strictly governed by the **Type of Room** and the **Location**. Our models captured these rules with over 82% accuracy, providing a clear map of the NYC rental economy.